



Case Study: Tana River Connectivity and Resiliency Project

Protecting lives and livelihoods through piloting Community Early Warning Systems (CEWS) and internet enablement in Tana River County, a climate disaster area in Kenya, supported by the Internet Society Foundation (ISOC).

Context: More than 370,000 people live in Tana River County, an Arid and Semi-Arid region that lies along the banks of the River Tana in Kenya. Kenya is made up of 47 counties, of which five have been declared climate disaster areas – Tana River is one of them, frequently experiencing flooding and drought.

The Challenge: There are many Community-Based Organizations (CBOs) running projects for their communities in Tana River County – but a lack of resources, unstable connectivity, and poor access to and understanding of available climate data means they find it difficult to pre-empt and prevent loss of life and livelihoods, and to prevent health crises like malnutrition, when a climate disaster happens.

Early Warning Systems (EWS) are proven to be one of the best and most cost-effective ways to reduce the impacts of natural disasters – having already saved tens of thousands of lives, and hundreds of billions of dollars ([UNDRR](#)). But as of 2022, less than half of developing countries were protected by multi-hazard EWS, meaning the people and communities most harshly and frequently affected by climate disasters don't have access to the early warning systems they need to respond to the threat.

Theory of Change: By training Tana River County CBOs and local government in EWS, and providing climate resilient internet connectivity, community leaders will have:

1. Improved infrastructure to access Early Warning Systems
2. Increased skills to interpret alerts
3. Enhanced mechanisms to transfer information to communities
4. Early preparedness and response capabilities
5. Opportunities to further develop their internet-enabled support to their communities

Enabling them to support their communities to save countless lives and livelihoods.

Intervention: With funding and support from the Internet Society Foundation (ISOC), NetHope piloted a localization project to provide 16 CBOs and local government actors from across Tana River County with training and resilient internet connectivity to access and



disseminate EWS. National systems already exist in Kenya to track climate trends and warn of disasters, so NetHope utilized these, working with early warning resources published by the Kenya Meteorological Department (KMD) and the National Drought Management Authority (NDMA) to equip community leaders in Tana River with the knowledge to interpret these warning systems.

NetHope created a multi-day, comprehensive *EWS and Connectivity Training* program that can be adapted to different contexts. The training covered basic digital skills as well as EWS, and is designed to be interactive, taking a 'learning by sharing and doing' approach to provide the participants with opportunities to practice, digest, and retain new knowledge and skills. NetHope recruited a Facilitator with deep knowledge of Tana River to further adapt the content to the local context and run it with the CBOs.

During the training, representatives from KMD and NDMA joined the CBO leaders and local government representatives to demystify existing EWS. Explaining which early warnings exist, how to access them, what the different thresholds are, and crucially, what the messages really mean for the people likely to be affected. NetHope's Facilitator for this project, Nyambura Wamaitha, said, **"That has been the most transformative thing about this NetHope project in the communities, because people now have the power to say, 'We know where to get the right kind of information'."**

By ensuring the government participated throughout the project, citizen engagement and support were ensured in the long term. For example, CBOs were connected to the county meteorological officer in case they need to ask further questions about climate data and impending weather events, and ensure that the officer would continue to include them in further adaptation and climate-related activities by the local government.

Included in the project, NetHope provided 16 Starlink kits to CBOs and the local government, and trained them on how to use and maintain them. The kits are climate resilient, easy to set up and set down, don't depend on physical infrastructure (that could be washed away by floods), and come with a solar panel to charge the battery when there is a long absence of electricity due to extreme weather events. These connectivity kits are not only a gateway to EWS for the communities, but are also being used to support the other work these community-based organizations undertake, for example, education, health, and women and children programs.

As a result of the project, CBOs have long-term skills and connectivity to access EWS and the internet, so they can receive the climate-related alerts, share the information with their communities in a way they'll be able to understand and act on (i.e. through social media messages, over the radio, in person through community meetings), and, crucially, mobilize their teams and their communities to respond early to an impending climate hazard or disaster.



Impact: The 16 trained CBOs and local government are already utilizing the internet connectivity and skills gained, and they report reaching over 200,000 people with appropriate climate hazard alerts. Within its project term (May 2024 to May 2025), CBOs had already shared stories that pointed to a reduction in the loss of lives and livelihoods.

The following quotes from participating CBOs give a snapshot of the ongoing impact of the internet access kits and digital training for the communities in Tana River.

- **"The comprehensive training from NetHope on early warning systems, complemented by the invaluable Starlink connectivity kit, has profoundly strengthened our capacity to serve the Tana Delta community."** – *Amani Center CBO*
- **"The training has also strengthened collaboration between our organization, county and national government agencies** since we partner during our activities. The training has also strengthened our actions and response planning, **which has led to more coordinated efficient climate [resilience] efforts"**. – *Tana County Pastoralist Forum*
- "The community rely on us to give them information. We already have stories that were successful. Abdulaziz, a young farmer, received a flood warning through WhatsApp. He quickly relocated from his vulnerable farm to higher ground. Using the rainwater, he planted tomatoes and harvested successfully. Abdulaziz said: **'The early warning I got through WhatsApp saved my livelihood. I moved my farm in time and used the rain to grow my tomatoes – I'm so grateful for this support.'**" – *Rama Bonaya, Founder of Tana Bora CBO, Tana River County*
- "We shared the alerts using social media and we also had a physical community meeting... **After they hearkened to the alerts people saved their property. They also saved their lives**, because normally these floods take life... But in this period of flash floods, after the alerts, there was no case of losing life." – *Yona Makonden, Chairman of Shakababo Community Conservancy, Tana River County*

The benefits of localization: This project has given organizations that are trusted and fully committed to supporting their communities access to information and resilient connectivity that can drastically reduce the impact of disasters. "CBOs now have the agency to come up with their response," said NetHope's Climate Intersection Lead, Gidraf Wachira. "What has happened most times in the past is that these communities have to wait for the government to do something... But in this case, communities can get the information early, understand it, and take timely protective action. They are also able to better interact with their government on how best to help them before, during, and after disaster."

"This is what communities need," Nyambura adds. "When you've been in a place like Tana River for decades and decades, aid has been the song of the lives of people. But **a project like this really empowers communities to take charge. To really sit down and understand deeply the problems and also the solutions – and after understanding, take small steps to build the solutions.**"



Next Steps: This pilot project was designed to be replicated, with training that can be easily adapted to local contexts and run by a facilitator using relevant local examples. The success of the pilot Connectivity and Resilience Project has shown that when CBOs have access to and understanding of EWS and the internet, then lives and livelihoods are better protected.

"Every community has its own climate change vulnerability, socio-economic indicators, and adaptation capacities, so adaptation has to be specific to a certain location," said Gidraf. "Adaptation also needs to be community-based, because communities are already experiencing these extreme weather events, and have already adapted in some ways. The local government needs to be active in these trainings because then they can be more responsive to what the community needs."

"What communities and their trusted CBOs need [are] partnerships and government commitments that accelerate their adaptive capacity, specifically with digital skills and digital resources to meet their immediate needs."

Having seen first-hand the impact of this project on her community, Facilitator Nyambura said: **"I think this kind of localization should be the thing that's embraced the most right now**, because you can really see how much difference it makes when the communities themselves are doing it, and because what it encourages more synergies from governments to communities."

Connectivity and Resilience in Tana River County, Kenya

Activities

Community-based organizations (CBOs) and local governments trained in Community Early Warning Systems (CEWS)

NetHope training on Kenya Meteorological Department (KMD) and National Drought Management Association (NDMA) early warning resources, in coordination with Tana River County Climate Change and Disaster Risk Reduction Steering Group



Donation of connectivity equipment and seed capital to support operation of the system

Trained CBOs equipped with climate resilient internet connectivity, including Starlink, solar powered battery and solar panel



Immediate impact

14 CBOs and local government from Tana River County (a highly climate vulnerable area) have increased capacity to provide early warnings to their communities

38 staff from CBOs and local government officials trained on CEWS and connectivity during climate emergencies

Long-term impact

Local communities from Tana River County (more than 200,000 people reached by CBOs) are more resilient and better prepared to respond during climate related shocks, protecting their lives and livelihoods

Through:

- Improved infrastructure to access Early Warning Systems
- Enhanced mechanisms to transfer information to communities
- Increased skills to interpret alerts
- Early preparedness and response



"Before the project, we were always the last to know about danger. Now, with the internet access and early warning alerts, our youth group can inform people quickly and even help plan responses. It has brought power and purpose to our community."

♥ Yusuf S
Youth CBO Member

"Through this [project], the county will effectively undertake disaster early warning dissemination of information to all community members on time by enhanced digital connectivity and improved IT infrastructure for better crisis management at the community level. This will not only empower our team but also improve service delivery to the community."

♥ Tana River County
Government

NETHOPE



By providing CBOs with training in CEWS and resilient communications equipment entire communities are equipped to save their livelihoods and lives



OFFICE OF THE PRESIDENT



COUNTY GOVERNMENT OF TANA RIVER

11th February 2025

To,
NetHope

Dear Sir/Madam

RE: APPRECIATION FOR SUPPORT AND REQUEST FOR CONNECTIVITY KIT

On behalf of both levels of Governments in Tana River County, the office express sincere appreciation for the invaluable support that NetHope has provided. Through implementing the Tana River Resilience and Connectivity Project, your commitment to leveraging technology for humanitarian impact will significantly enhance our ability to serve communities in need. The resources, expertise, and collaboration extended to will be very instrumental in advancing our mission, and we are truly grateful for the partnership.

Through this, the county will effectively undertake disaster early warning dissemination of information to all community members on time by enhanced digital connectivity and improved IT infrastructure for better crisis management at the community level. This will not only empower our team but also improved service delivery to the community.

As we continue to coordinate these efforts, we kindly request NetHope to consider expanding this support to the county government specifically the county commissioner and the special programs department by providing them with the connectivity kits. The two departments are mandated to coordinate and run all Disaster Management issues in the county. The above mentioned device will assist the County to better manage disaster response effectively and timely.

We strongly believe that with NetHope's continued partnership, we can further scale our impact and bridge critical gaps in the disaster management sector especial in early warning information dissemination. We would appreciate the opportunity to discuss how we can collaborate further to achieve our shared goals.

We thank you for your continued support.

HALUBVA ANNAH GWIYO
AG: ASSISTANT DIRECTOR
COHESION & SPECIAL PROGRAMS
COUNTY GOVERNMENT OF TANA RIVER